Filippo Maggioli, Ph.D.

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Short Bio

- **Research profile**
  I am a Tenured Assistant Professor at *Pegaso University*. Previously, I was Postdoctoral Researcher at *University of Milano-Bicocca* in the *DIG AIR* research lab led by Simone Melzi, a Postdoctoral Researcher at *Sapienza – University of Rome* in the *GLADIA* research lab led by Emanuele Rodolá, and a Research Intern at the *King Abdullah University of Science and Technology (KAUST)* in the *VCC* research lab led by Peter Wonka. I received my Ph.D. in Computer Science at *Sapienza – University of Rome* (2023), where I also graduated in Computer Science (2019).
  I work on geometry processing, spectral geometry, and 3D shape analysis, but I am an active researcher also in other fields of computer graphics, such as procedural shading and physical simulation.
  I regularly serve in the program committee of international conferences as chair and reviewer, and I maintain worldwide collaborations with researchers from other institutions and countries.
- **Research interests**
  Geometry Processing; Spectral Geometry; 3D Shape Analysis; Procedural Texturing; Simulation of Natural Phenomena; Numerical Linear Algebra.
- **Author profiles**

ORCID:	0000-0001-8008-8468		
Google Scholar ID:	VN1fbwUAAAAJ	*h*-index: 7	*i10*-index: 5
Scopus Author ID:	57216313662	*h*-index: 5	*i10*-index: 3

Academic Appointments

Mar 2025 – Present	**Assistant professor - tenure track.** Pegaso University Department of Computer Science and Information Technologies.
Dec 2023 – Feb 2025	**Postdoctoral researcher.** University of Milano-Bicocca Member of the *DIG AIR* research lab. Advisor: prof. Simone Melzi. Research activity on computational and spectral geometry.
Apr 2024 – Feb 2025	**Adjunct professor.** Pegaso University Undergrad courses on *Computer Achitecture* and *Networking and Cybersecurity*. Supervision of BSc students during the development of their theses.
Aug 2023 – Nov 2023	**Postdoctoral researcher.** Sapienza – University of Rome Member of the *Gladia* research lab. Advisor: prof. Emanuele Rodolà. Research activity on computational geometry, spectral geometry, and numerical linear algebra.
Sep 2022 – Jan 2023	**Research internship.** King Abdullah University of Science and Technology Member of the *VCC* research lab. Advisor: prof. Dominik L. Michels. Research activity on simulation of natural phenomena in agricultural settings.
Mar 2021 – Jul 2021	**Teaching assistant.** Sapienza – University of Rome Undergrad course on *Introduction to Algorithms*.

Education

2019 – 2023 **Ph.D. in Computer Science.** Sapienza – University of Rome.
Advisor: prof. Emanuele Rodolà.
Thesis title: *Scalable geometry processing for computer graphics applications.*
Honourable mention at EG-Italy Award for PhD Thesis in Computer Graphics.

2018 – 2019 **M.Sc. in Computer Science.** Sapienza – University of Rome.
Advisor: prof. Emanuele Rodolà.
Thesis title: *Time-efficient function reconstruction via Laplacian eigenproducts.*

2014 – 2017 **B.Sc. in Computer Science.** Sapienza – University of Rome.
Advisor: prof. Enrico Tronci.
Thesis title: *Modeling of biological pathways with systems of differential-algebraic equations.*

Courses and Schools

Jul 2022 **IRDTA DeepLearn 2022 Summer.**
6th International Gran Canaria School on Deep Learning. *Las Palmas de Gran Canaria, Spain.*

Jul 2021 **ACDL 2021.**
Advanced Online & Onsite Course on Data Science & Machine Learning. *Pontignano, Italy.*

Academic Service

2026 – Present **Associate editor.** EJ-AI
European Journal on Artificial Intelligence.

2026 **Guest editor.** Graphical Models
Virtual Special Issue on Smart Tools and Applications in Graphics 2026.

Program chair. STAG
Smart Tools and Applications in Graphics. *Catania, Italy*

Member of program committee. SGP
Symposium on Geometry Processing. *Bern, Switzerland*

Member of program committee. AICCSA
ACS/IEEE International Conference on Computer Systems and Applications. *Sharjah, UAE*

2025 **Poster chair.** STAG
Smart Tools and Applications in Graphics. *Genova, Italy*

Member of program committee. AICCSA
ACS/IEEE International Conference on Computer Systems and Applications. *Doha, Qatar*

Member of program committee. ECAI
European Conference on Artificial Intelligence. *Bologna, Italy*

2023 **Member of program committee.** TAG-ML
ICML's workshop on Topology, Algebra, and Geometry in Machine Learning. *Honolulu, Hawaii*

2022 **Member of RCDC Conference Coffee Committee.** ACM SIGGRAPH RCDC
ACM SIGGRAPH Research Career Development Committee. *Vancouver, Canada*

2021 **Event chair.** STAG
Smart Tools and Applications in Graphics. *Rome, Italy*

Teaching

2026 — **The Ubiquity of Voronoi Diagrams**
Workshop (4 hours). *Central European Seminar on Computer Graphics (CESCG)*

Innovative Methods, Gamification, and Digital Applications
Excellence School (4 hours). *Pegaso University*

Teaching Networking
Teaching qualification training course (9 hours). *Pegaso University*

2025 – Present — **Algorithms and Data Structures**
Undergrad course (30 hours/year). *Pegaso University*

Networking and Cybersecurity
Undergrad course (30 hours/year). *Pegaso University*

Web Technologies
Undergrad course (30 hours/year). *Pegaso University*

2025 — **Functional Maps: a spectral approach to the alignment of embeddings**
Ph.D. course (12 hours). *Sapienza – University of Rome*

Digital Skills for Teaching
Teaching qualification training course (12 hours). *Pegaso University*

Teaching Networking
Teaching qualification training course (18 hours). *Pegaso University*

2024 – 2025 — **Networking and Cybersecurity**
Undergrad course (60 hours/year). *Pegaso University*

Computer Architecture
Undergrad course (30 hours/year). *Pegaso University*

2021 — **Introduction to Algorithms**
Undergrad course (30 hours). *Sapienza – University of Rome*

Invited, Conference, and Seminars Talks

Nov 2025 — **UV Parametrization via Topological Disk Segmentation of Surfaces**
Smart Tools and Applications in Graphics, 2025. *Genoa, Italy*

Jun 2025 — **Volumetric Functional Maps**
Sapienza – University of Rome, hosted by *E. Rodolà*. *Rome, Italy*.

Nov 2024 — **Scalable geometry processing in computer graphics applications**
Smart Tools and Applications in Graphics, 2024. *Verona, Italy*

TACO: a benchmark for connectivity-invariance in shape correspondence
Smart Tools and Applications in Graphics, 2024. *Verona, Italy*

Efficient Generation of Multimodal Fluid Simulation Data
Smart Tools and Applications in Graphics, 2024. *Verona, Italy*

Dec 2023 — **A physically-inspired approach to the simulation of plant wilting**
ACM SIGGRAPH Asia, 2023. *Sydney, Australia*.

Oct 2022 — **MoMaS: mold manifold simulation for real-time procedural texturing**
Pacific Graphics (PG), 2022. *Kyoto, Japan*.

May 2022 — **Strassen's algorithm in practice**
Sapienza – University of Rome, hosted by *R. Marin*. *Rome, Italy*.

Aug 2021 — **Efficiently parallelizable Strassen-based multiplication of a matrix by its transpose**
International Conference on Parallel Processing (ICPP), 2021. *Chicago, Illinois, USA*.

May 2021 — **Orthogonalized Fourier polynomials for signal approximation and transfer**
EUROGRAPHICS (EG), 2021. *Vienna, Austria*.

Honours & Awards

2026 **CVPR 2026 Outstanding Reviewer**

2025 **STAG Best Paper Award – Honourable Mention**
UV Parametrization via Topological Disk Segmentation of Surfaces
Smart Tools and Applications in Graphics.

2024 **Matteo Dellepiane Award for PhD Thesis in Computer Graphics – Honourable Mention**
The Italian Chapter of EuroGraphics (EG-Italy).

Research Grants and Funding

2022 **Sapienza Research Starting Grant: Avvio alla Ricerca – Tipo 2**
Principal investigator for the project *Enhancing Procedural Computer Graphics in Multimedia Applications with Fast Geometry Processing Techniques*.

2021 **Sapienza Research Starting Grant: Avvio alla Ricerca – Tipo 1**
Principal investigator for the project *Automation of Casting Mold Design for Industrial Fabrication of Digital Objects*.

2020 **Sapienza Research Starting Grant: Avvio alla Ricerca – Tipo 1**
Principal investigator for the project *GPU Fluid Simulation on Non-Euclidean Domains and Application for Simulation of Erosion Phenomena*.

Reviewing Service

2026 SIGGRAPH, SIGGRAPH Asia, NeurIPS, JOCCH, ECCV, JMIV, BMVC, SGP, TVCG, CVPR

2025 Pacific Graphics, ECAI, GCA, ICCV, CVPR, TPAMI, CAVW, EUROGRAPHICS

2024 ToG, ACCV, Pacific Graphics, ECCV, EUROGRAPHICS, CGF, TVCG

2023 Pacific Graphics, TAG-ML, ICCV, NeurReps, ICIAP

2022 EUROGRAPHICS

Supervision and Mentoring

Ph.D. Giulio Viganó (University of Milano-Bicocca) – Internal supervisor (not formal advisor)
Francesca Maccarone (University of Milano-Bicocca) – Internal supervisor (not formal advisor)
Daniele Baieri (Sapienza – University of Rome) – Internal supervisor (not formal advisor)
Francesco De Canio (Sapienza – University of Rome) – Internal supervisor (not formal advisor)

M.Sc. Giorgio Longari (University of Milano-Bicocca) – Internal supervisor (not formal advisor)
Roberta Giorgi (Sapienza – University of Rome) – Internal supervisor (not formal advisor)
Thesis advisor of 6 students (Pegaso University)

B.Sc. Alessandro Sommariva (University of Milano-Bicocca) – Thesis co-advisor
Simone Pedico (University of Milano-Bicocca) – Thesis co-advisor
Alessio Tosato (University of Milano-Bicocca) – Thesis co-advisor
Arianna Silvestri (University of Milano-Bicocca) – Internal supervisor (not formal advisor)
Alireza Alipanah (Sharif University of Technology) – Internal supervisor (not formal advisor)
Daniele Solombrino (Sapienza – University of Rome) – Internal supervisor (not formal advisor)
Thesis advisor of 11 students (Pegaso University)

Skills

Languages	Italian (mother tongue), English (professional proficiency).
Interpersonal	Adaptability to work independently and with(in) a team. Capability of supervising and communicating efficaciously. Excellent organizational and teaching abilities.
Programming	Proficient in C/C++ and MATLAB. Advanced knowledge of GPU programming with CUDA, GLSL, and HLSL. Knowledge of Python and C#.
Tools	Expert with the mesh processing software *MeshLab* and the rendering engine *Blender*. Advanced knowledge of the game engines *Unreal Engine 4 and 5* and *Unity 3D*. Familiar with software for raster (*GIMP*) and vector (*InkScape*) 2D graphics.

Research Publications

Journal Articles

1. D. Marin, **F. Maggioli**, S. Melzi, S. Ohrhallinger, and M. Wimmer, "Reconstructing curves from sparse samples on riemannian manifolds," *Computer Graphics Forum*, vol. 43, no. 5, e15136, 2024.
2. **F. Maggioli**, R. Marin, S. Melzi, and E. Rodolà, "Momas: Mold manifold simulation for real-time procedural texturing," *Computer Graphics Forum*, vol. 41, no. 7, pp. 519–527, 2022.
3. L. Moschella, S. Melzi, L. Cosmo, **F. Maggioli**, O. Litany, M. Ovsjanikov, L. Guibas, and E. Rodolà, "Learning spectral unions of partial deformable 3d shapes," *Computer Graphics Forum*, vol. 41, no. 2, pp. 407–417, 2022.
4. **F. Maggioli**, S. Melzi, M. Ovsjanikov, M. M. Bronstein, and E. Rodolà, "Orthogonalized fourier polynomials for signal approximation and transfer," *Computer Graphics Forum*, vol. 40, no. 2, pp. 435–447, 2021.
5. **F. Maggioli**, T. Mancini, and E. Tronci, "Sbml2modelica: Integrating biochemical models within open-standard simulation ecosystems," *Bioinformatics*, vol. 36, no. 7, pp. 2165–2172, 2020.

Conference Proceedings

1. **F. Maggioli**, V. De Luca, A. Generosi, L. Galteri, and L. Gallo, "Distance-based iterated function systems on surfaces," in *ACM SIGGRAPH 2026 Posters*, 2026.
2. **F. Maggioli**, M. Livesu, and S. Melzi, "Volumetric functional maps," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.
3. L. Olearo, G. Viganò, **F. Maggioli**, D. Baieri, and S. Melzi, "Fuse: A flow-based mapping between shapes," in *Computer Vision – ECCV 2026*, 2026.
4. D. Baieri, **F. Maggioli**, E. Rodolà, S. Melzi, and Z. Lähner, "Implicit-arap: Efficient handle-guided neural field deformation via local patch meshing," in *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
5. A. Ferraris, F. Leveni, D. Baieri, **F. Maggioli**, S. Melzi, and L. Magri, "Geometric aware local optimization for robust primitive fitting," in *Smart Tools and Applications in Graphics-Eurographics Italian Chapter Conference*, 2025.
6. **F. Maggioli**, D. Baieri, E. Rodolà, and S. Melzi, "Rematching: Low-resolution representations for scalable shape correspondence," in *Computer Vision – ECCV 2024*, Cham: Springer Nature Switzerland, 2025, pp. 183–200, ISBN: 978-3-031-72913-3.
7. **F. Maggioli** and S. Melzi, "Uv parametrization via topological disk segmentation of surfaces," in *Smart Tools and Applications in Graphics-Eurographics Italian Chapter Conference*, 2025.

8. D. Baieri, D. Crisostomi, S. Esposito, **F. Maggioli**, and E. Rodolà, "Efficient generation of multimodal fluid simulation data," in *Smart Tools and Applications in Graphics-Eurographics Italian Chapter Conference*, 2024.
9. F. Maccarone, G. Longari, G. Viganò, D. Peruzzo, **F. Maggioli**, and S. Melzi, "S4a: Scalable spectral statistical shape analysis," in *Smart Tools and Applications in Graphics-Eurographics Italian Chapter Conference*, 2024.
10. S. Pedico, S. Melzi, and **F. Maggioli**, "Taco: A benchmark for connectivity-invariance in shape correspondence," in *Smart Tools and Applications in Graphics-Eurographics Italian Chapter Conference*, 2024.
11. **F. Maggioli**, J. Klein, T. Hädrich, E. Rodolà, W. Pałubicki, S. Pirk, and D. L. Michels, "A physically-inspired approach to the simulation of plant wilting," in *SIGGRAPH Asia 2023 Conference Papers*, 2023, pp. 1–8.
12. **F. Maggioli**, D. Baieri, S. Melzi, and E. Rodolà, "Newton's fractals on surfaces via bicomplex algebra," in *ACM SIGGRAPH 2022 Posters*, 2022, pp. 1–2.
13. V. Arrigoni, **F. Maggioli**, A. Massini, and E. Rodolà, "Efficiently parallelizable strassen-based multiplication of a matrix by its transpose," in *Proceedings of the 50th International Conference on Parallel Processing*, 2021, pp. 1–12.

Pre-prints

1. **F. Maggioli**, D. Baieri, Z. Lähner, and S. Melzi, "Sshade: A framework for scalable shape deformation via local representations," arXiv preprint arXiv:2409.17961, 2024.
2. D. Baieri, S. Esposito, **F. Maggioli**, and E. Rodolà, "Fluid dynamics network: Topology-agnostic 4d reconstruction via fluid dynamics priors," arXiv preprint arXiv:2303.09871, 2023.